

0.1 Testing Electronics

2	0	1	0	0	1	37	512	473	Pos: 7404
2	1	1	0	1	3	39	512	469	Pos: 7385
2	1	1	0	1	1	39	518	468	Pos: 7387
2	0	3	0	0	1	39	507	465	Pos: 7388
2	0	0	2	5	1	39	506	469	Pos: 7381
2	0	0	4	1	0	39	511	466	Pos: 7385
2	0	0	0	0	1	39	518	466	Pos: 7400
2	1	1	0	1	0	39	519	466	Pos: 7387
2	0	0	10	1	1	37	511	466	Pos: 7359
2	0	1	0	0	1	39	511	463	Pos: 7396
2	1	1	0	1	1	37	512	466	Pos: 7390
2	0	1	0	0	1	37	511	466	Pos: 7400
2	0	1	2	0	1	39	512	465	Pos: 7388
2	3	1	0	1	1	39	518	469	Pos: 7375
2	0	1	2	0	0	37	509	471	Pos: 7398
2	1	1	0	0	0	39	518	473	Pos: 7395

Figure 1: Description of IR sensor

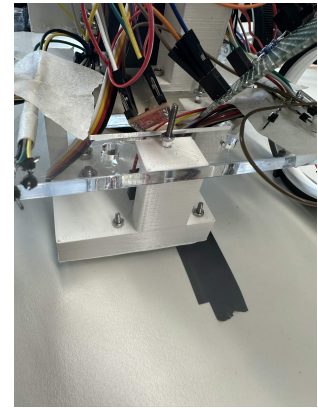


Figure 2: Description of line sensor

5	5	67	632	599	668	497	6	3	Pos: 4358
6	3	65	639	573	668	499	15	4	Pos: 4374
7	5	58	639	585	680	502	13	4	Pos: 4377
6	5	65	639	589	672	504	13	4	Pos: 4371
6	5	65	632	575	678	504	12	0	Pos: 4371
6	5	63	643	587	660	502	13	7	Pos: 4371
9	5	63	634	585	678	500	13	6	Pos: 4373
5	5	65	634	585	668	499	15	4	Pos: 4374
6	3	65	634	575	672	497	13	4	Pos: 4374
6	10	65	632	573	668	508	15	3	Pos: 4372
6	3	65	632	573	670	502	15	4	Pos: 4380
6	5	65	628	575	670	499	15	4	Pos: 4377
6	3	65	645	579	678	504	13	4	Pos: 4373
6	5	65	639	579	672	509	13	4	Pos: 4376
6	0	65	643	583	680	500	13	4	Pos: 4376
6	3	65	641	581	670	502	13	4	Pos: 4372

Figure 3: Description of IR sensor

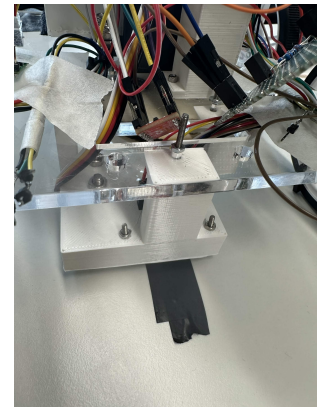


Figure 4: Description of line sensor

561	577	253	4	1	3	1	3	1	Pos: 818
565	577	253	2	1	3	3	3	1	Pos: 820
565	575	259	6	1	3	1	3	3	Pos: 834
567	577	253	6	1	3	1	3	1	Pos: 817
565	579	255	8	1	3	0	3	3	Pos: 830
564	589	255	4	1	3	1	3	1	Pos: 819
564	588	253	8	1	3	0	1	6	Pos: 837
563	588	253	2	1	3	0	5	4	Pos: 835
564	579	255	10	1	1	0	3	3	Pos: 828
565	589	251	4	1	3	1	3	3	Pos: 826
561	568	253	6	0	3	1	1	1	Pos: 809
566	579	259	6	3	3	0	3	3	Pos: 834
564	588	257	6	1	3	1	5	3	Pos: 843
564	588	257	6	1	3	1	3	3	Pos: 834
566	580	255	6	3	3	0	5	3	Pos: 840

Figure 5: Description of IR sensor

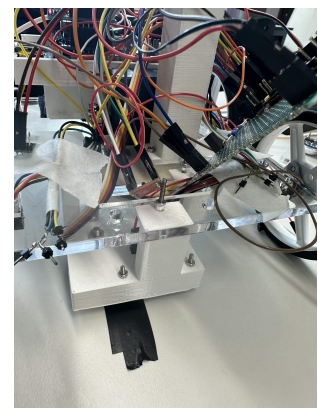


Figure 6: Description of line sensor

2	1	1	0	0	1	0	0	1	Pos: 0
3	1	1	2	0	1	0	1	0	Pos: 0
2	0	1	0	0	1	0	1	0	Pos: 0
2	1	1	0	0	1	0	1	0	Pos: 0
2	1	1	2	0	0	0	0	0	Pos: 0
2	0	0	0	0	1	0	3	0	Pos: 0
2	1	1	0	0	0	0	1	1	Pos: 0
2	1	0	0	0	1	0	0	1	Pos: 0
2	0	1	4	0	1	0	0	0	Pos: 0
2	1	1	0	0	1	0	0	0	Pos: 0
2	1	1	2	0	1	0	0	0	Pos: 0
2	1	0	0	1	1	0	1	0	Pos: 0
2	0	0	0	0	1	0	0	3	Pos: 0
2	1	1	0	1	1	0	0	0	Pos: 0
3	1	1	0	0	1	0	0	0	Pos: 0
2	0	0	10	0	1	0	1	0	Pos: 0

Figure 7: Description of IR sensor

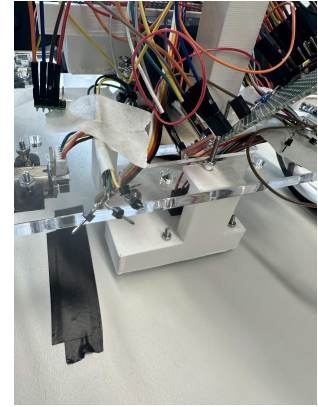
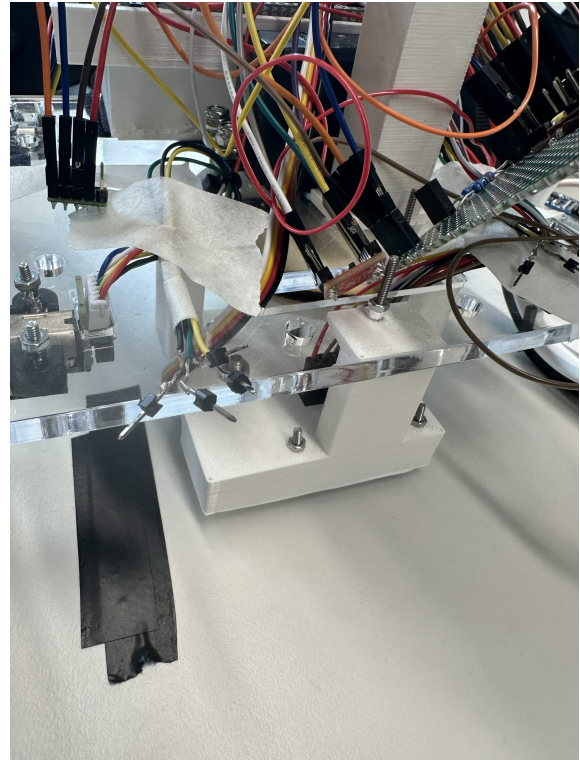


Figure 8: Description of line sensor

Figure 9–12 show the sensor readings and corresponding robot position for four line configurations.

2	1	1	0	0	1	0	0	1	Pos: 0
3	1	1	2	0	1	0	1	0	Pos: 0
2	0	1	0	0	1	0	1	0	Pos: 0
2	1	1	0	0	1	0	1	0	Pos: 0
2	1	1	2	0	0	0	0	0	Pos: 0
2	0	0	0	0	1	0	3	0	Pos: 0
2	1	1	0	0	0	0	1	1	Pos: 0
2	1	0	0	0	1	0	0	1	Pos: 0
2	0	1	4	0	1	0	0	0	Pos: 0
2	1	1	0	0	1	0	0	0	Pos: 0
2	1	1	2	0	1	0	0	0	Pos: 0
2	1	0	0	1	1	0	1	0	Pos: 0
2	0	0	0	1	0	0	0	3	Pos: 0
2	1	1	0	1	1	0	0	0	Pos: 0
3	1	1	0	0	1	0	0	0	Pos: 0
2	0	0	10	0	1	0	1	0	Pos: 0

(a) Serial Monitor readings when off the line.

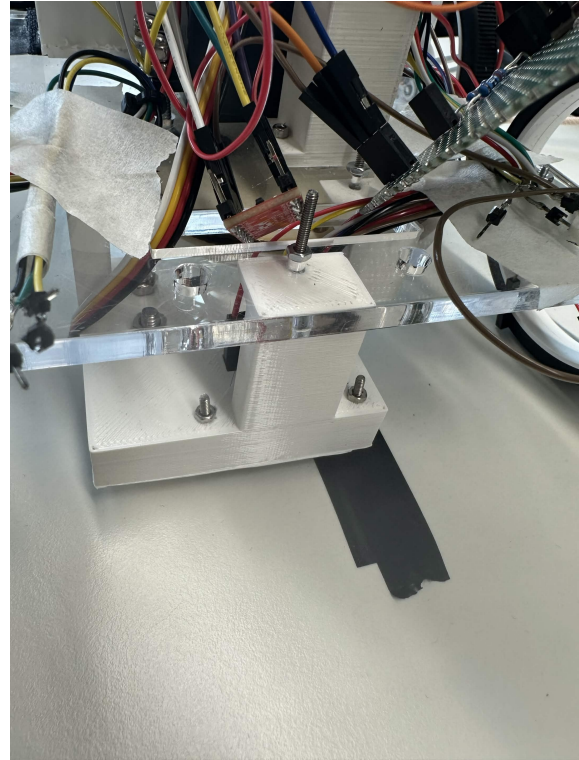


(b) Robot position when off the line.

Figure 9: Reflectance sensor readings and robot position: off the line.

2	0	1	0	0	1	37	512	473	Pos: 7404
2	1	1	0	1	3	39	512	469	Pos: 7385
2	1	1	0	1	1	39	518	468	Pos: 7387
2	0	3	0	0	1	39	507	465	Pos: 7388
2	0	0	2	5	1	39	506	469	Pos: 7381
2	0	0	4	1	0	39	511	466	Pos: 7385
2	0	0	0	0	1	39	518	466	Pos: 7400
2	1	1	0	1	0	39	519	466	Pos: 7387
2	0	0	10	1	1	37	511	466	Pos: 7359
2	0	1	0	0	1	39	511	463	Pos: 7396
2	1	1	0	1	1	37	512	466	Pos: 7390
2	0	1	0	0	1	37	511	466	Pos: 7400
2	0	1	2	0	1	39	512	465	Pos: 7388
2	3	1	0	1	1	39	518	469	Pos: 7375
2	0	1	2	0	0	37	509	471	Pos: 7398
2	1	1	0	0	0	39	518	473	Pos: 7395

(a) Serial Monitor readings when left of the line.

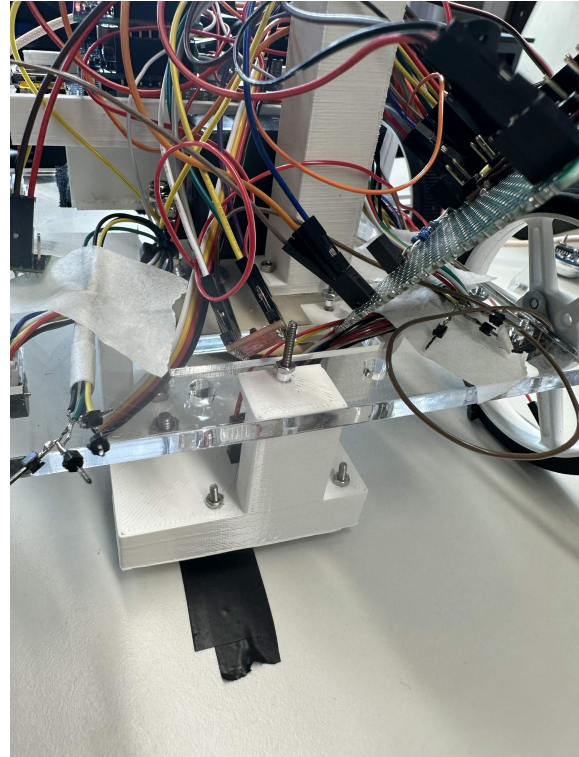


(b) Robot position when left of the line.

Figure 10: Reflectance sensor readings and robot position: left of the line.

561	577	253	4	1	3	1	3	1	Pos: 818
565	577	253	2	1	3	3	3	1	Pos: 820
565	575	259	6	1	3	1	3	3	Pos: 834
567	577	253	6	1	3	1	3	1	Pos: 817
565	579	255	8	1	3	0	3	3	Pos: 830
564	589	255	4	1	3	1	3	1	Pos: 819
564	588	253	8	1	3	0	1	6	Pos: 837
563	588	253	2	1	3	0	5	4	Pos: 835
564	579	255	10	1	1	0	3	3	Pos: 828
565	589	251	4	1	3	1	3	3	Pos: 826
561	568	253	6	0	3	1	1	1	Pos: 809
566	579	259	6	3	3	0	3	3	Pos: 834
564	588	257	6	1	3	1	5	3	Pos: 843
564	588	257	6	1	3	1	3	3	Pos: 834
566	580	255	6	3	3	0	5	3	Pos: 840

(a) Serial Monitor readings when right of the line.

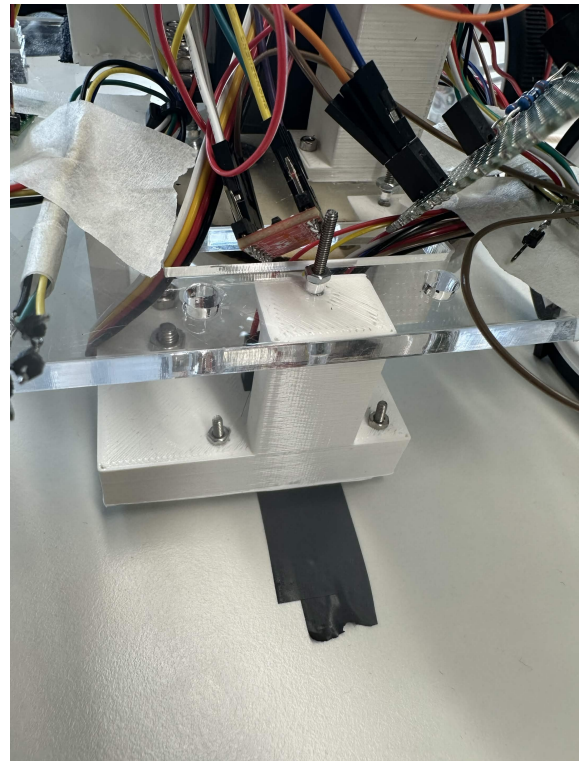


(b) Robot position when right of the line.

Figure 11: Reflectance sensor readings and robot position: right of the line.

5	5	67	632	599	668	497	6	3	Pos: 4358
6	3	65	639	573	668	499	15	4	Pos: 4374
7	5	58	639	585	680	502	13	4	Pos: 4377
6	5	65	639	589	672	504	13	4	Pos: 4371
6	5	65	632	575	678	504	12	0	Pos: 4371
6	5	63	643	587	660	502	13	7	Pos: 4371
9	5	63	634	585	678	500	13	6	Pos: 4373
5	5	65	634	585	668	499	15	4	Pos: 4374
6	3	65	634	575	672	497	13	4	Pos: 4374
6	10	65	632	573	668	508	15	3	Pos: 4372
6	3	65	632	573	670	502	15	4	Pos: 4380
6	5	65	628	575	670	499	15	4	Pos: 4377
6	3	65	645	579	678	504	13	4	Pos: 4373
6	5	65	639	579	672	509	13	4	Pos: 4376
6	0	65	643	583	680	500	13	4	Pos: 4376
6	3	65	641	581	670	502	13	4	Pos: 4372

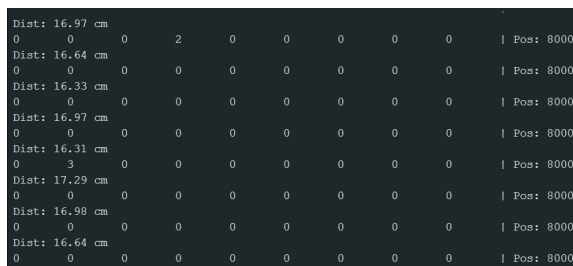
(a) Serial Monitor readings when centred on the line.



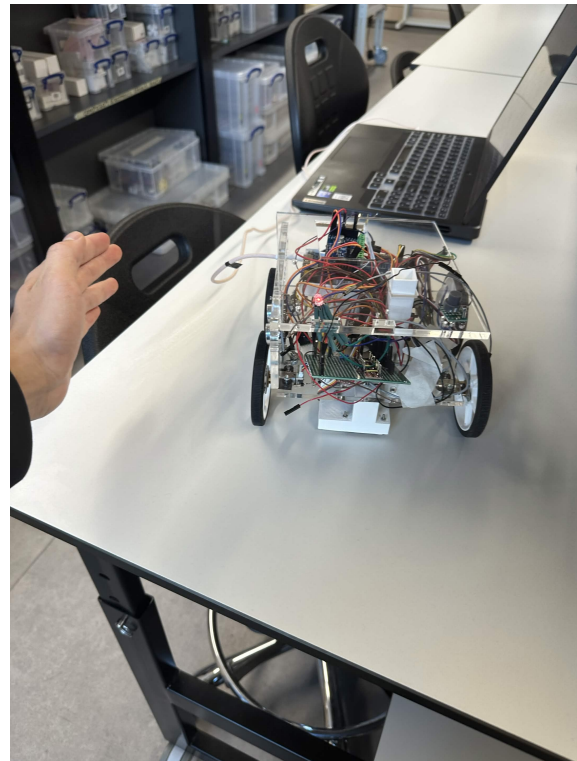
(b) Robot position when centred on the line.

Figure 12: Reflectance sensor readings and robot position: centred on the line.

Figure 13 shows the HC-SR04 waveforms and a demonstration of live distance readings.



(a) HC-SR04 trigger and echo signal waveforms.



(b) Live distance readings from the sensor.

Figure 13: HC-SR04 ultrasonic sensor signals and distance measurement demo.

```

16:24:07.786 -> StateMachine =
16:24:07.822 -> Card UID: 7074AB41
16:24:07.857 -> [SENT] type=isFertile tag_id=7074AB41
16:24:07.974 -> MSG: type=isFertileReply tag_id=7074AB41 fertile=true planted=false x=2 y=8
16:24:07.974 -> === isFertileReply ===
16:24:07.974 -> Tag: 7074AB41
16:24:07.974 -> Fertile: true
16:24:07.974 -> Planted: false
16:24:07.974 -> Pos: x=2 y=8
16:24:07.974 -> RFID: fertile and unplanted - planting!
16:24:08.605 -> MSG:
16:24:09.082 -> [SENT] type=register team_id=1 board_id=Master
16:24:09.082 -> Registered.
16:24:09.159 -> [SENT] type=planted tag_id=7074AB41
16:24:09.159 -> RFID: seeds remaining: 3
16:24:09.197 -> StateMachine =

```

(a) Serial debug messages about the planting mechanism.

```
>>> Power button - isEnabled now true
[NAV] re-engaged - resuming state=NAV_BASE_TO_FIRST_JUNCTION
[NAV] -> NAV_BASE_TO_FIRST_JUNCTION :: following line toward first junction
MSG:
[BASE] first junction reached
[NAV] -> NAV_BASE_FIRST_TURN :: turning at first junction
[BASE] first turn 90.00 deg
[BASE] pre-turn nudge 3.00 cm to centre on junction
MSG:
MSG:
Turn complete.
[NAV] -> NAV_BASE_TO_TAG :: following line toward RFID tag
MSG:
[RECOVER] forward 3.00 cm then  $\pm 90^\circ$  sweep
MSG:
[SENT] type=register team_id=1 board_id=Master
Registered.
MSG:
[RECOVER] line acquired on left sweep
[RECOVER] forward 3.00 cm then  $\pm 90^\circ$  sweep
MSG:
[RECOVER] line acquired on left sweep
MSG:
[BASE] tag detected: C2834BF4
[BASE] sending openAirlockA - waiting for exitClearance before continuing
[SENT] type=openAirlockA board_id=Master
[NAV] -> NAV_WAIT_EXIT_CLEARANCE :: holding for exitClearance
MSG: type=openAirlockReply airlock=A accepted=false reason=missing_tag
>>> openAirlockReply A REJECTED queue_enter= queue_exit=
MSG:
MSG:
MSG:
[BASE] no exitClearance yet - resending openAirlockA
[SENT] type=openAirlockA board_id=Master
MSG: type=openAirlockReply airlock=A accepted=false reason=missing_tag
>>> openAirlockReply A REJECTED queue_enter= queue_exit=
MSG:
MSG:
MSG:
[BASE] no exitClearance yet - resending openAirlockA
[SENT] type=openAirlockA board_id=Master
MSG: type=openAirlockReply airlock=A accepted=false reason=missing_tag
>>> openAirlockReply A REJECTED queue_enter= queue_exit=
```

(a) Serial debug messages for navigating the base and exiting.